



LINUX SECURITY CHEAT SHEET

```
$ whoami  
$ ip a  
$ ss -tulpen  
$ sudo apt update  
$ tail -f /var/log/auth.log  
$ ufw status  
$ systemctl status ssh  
...
```



CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

1. USER & ACCOUNT SECURITY



	List users	<pre>\$ cat /etc/passwd</pre>	Lists all user accounts on the system.
	Check Logged-in Users	<pre>\$ who \$ w \$ users</pre>	Shows currently logged-in users.
	Last Login History	<pre>\$ last \$ lastlog</pre>	Shows last login history of users.
	Create User	<pre>\$ sudo adduser username</pre>	Creates a new user account.
	Delete User	<pre>\$ passwd username</pre>	Deletes a user account.
	Lock User Account	<pre>\$ passwd -l username</pre>	Locks a user account to prevent login.
	Unlock User Account	<pre>\$ passwd -u username</pre>	Unlocks a user account to allow login.
	Check Sudo Users	<pre>\$ cat /etc/sudoers \$ sudo -l</pre>	Lists users with sudo privileges.



WHY IT MATTERS

Strong user & account management helps prevent unauthorized access and privilege misuse.

★ TIP

Regularly review user accounts and remove inactive users.

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ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

2. PASSWORD & AUTHENTICATION SECURITY



Check Password Policy

```
$ cat /etc/login.defs
```

View system-wide password policy settings.



Check Password Expiry

```
$ chage -l username
```

Shows password aging and expiry information for a user.



Check Failed Login Attempts

```
$ grep "Failed password" \
/var/log/auth.log
```

Find failed SSH or login attempts.



Check PAM Configuration

```
$ cat /etc/pam.d/common-auth
```

View PAM (Pluggable Authentication Modules) configuration.



Check Root Login Status

```
$ grep PermitRootLogin \
/etc/ssh/sshd_config
```

Check if root login via SSH is allowed.



Check Account Lockout Policy

```
$ cat /etc/security/faillock.conf
```

View account lockout settings for failed login attempts.



Force Password Change on Next Login

```
$ passwd -e username
```

Forces user to change password at next login.



Check Account Aging Information

```
$ chage -l username
```

Displays account expiration and password aging details.



WHY IT MATTERS

Strong authentication controls reduce the risk of unauthorized access and brute force attacks.



TIP

Regularly review password policies and monitor failed login attempts to detect suspicious activity early.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

3. FILE PERMISSION SECURITY



Check File Permissions

```
$ ls -l
```

Displays permission details of files and directories.



Find SUID Files

```
$ find / -type f -perm -4000 -ls 2>/dev/null
```

Finds files with SUID permissions.



Find SGID Files

```
$ find / -type f -perm -2000 -ls 2>/dev/null
```

Finds files with SGID permissions.



Check World-writable Files

```
$ find / -type f -perm -0002 -ls 2>/dev/null
```

Finds files that are world-writable.



Check World-writable Directories

```
$ find / -type d -perm -0002 -ls 2>/dev/null
```

Finds directories that are world-writable.



Check File Ownership

```
$ stat filename
```

Shows detailed file ownership information.



Check File Capabilities

```
$ getcap -r / 2>/dev/null
```

Lists files with Linux capabilities set.



Check ACL Permissions

```
$ getfacl filename
```

Displays Access Control List (ACL) permissions.



Set Secure Permissions

```
$ chmod 600 filename
```

Restricts file access (owner read/write only).



Secure Directory Permissions

```
$ chmod 750 directory
```

Sets secure permissions for directories.



WHY IT MATTERS

Incorrect file permissions can lead to privilege escalation, data leakage, and unauthorized changes. Regular checks help prevent exploitation.



TIP

Always review SUID, SGID, and world-writable files. Attackers often exploit misconfigured permissions to gain access.

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ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

4. PROCESS MONITORING



	Running Processes	<code>\$ ps aux</code>	Lists all currently running processes.
	Interactive Process Monitor	<code>\$ top</code>	Real-time interactive process monitor.
	Better Tool (Interactive)	<code>\$ htop</code>	Enhanced interactive process viewer.
	Kill Process	<code>\$ kill PID</code>	Terminates a process by Process ID (PID).
	Force Kill Process	<code>\$ kill -9 PID</code>	Forcefully terminates a process by PID.
	Search Process by Name	<code>\$ ps aux grep process_name</code>	Finds a process by name.
	Show Process Tree	<code>\$ pstree</code>	Displays processes in a tree (parent-child) view.
	Check Process Start Time	<code>\$ ps -eo pid,comm,lstart</code>	Shows start time of running processes.
	Show Detailed Process Info	<code>\$ ps -fp PID</code>	Displays detailed information about a specific process.
	Monitor Process in Real-time	<code>\$ watch -n 1 ps aux</code>	Continuously monitors processes every second.



WHY IT MATTERS

Monitoring processes helps detect suspicious or malicious activity, resource abuse, and unauthorized execution.



TIP

Always investigate unknown or high CPU/memory processes. Attackers often use legitimate process names to hide malicious activity.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

5. NETWORK SECURITY



Check Open Ports

```
$ netstat -tulnp
or
$ ss -tulnp
```

Lists all open listening ports and associated processes.



Check Listening Services

```
$ lsof -i
```

Shows processes listening on network ports.



Active Connections

```
$ netstat -antp
```

Displays all active network connections.



Check Firewall Status

```
$ sudo ufw status
```

Checks the status and rules of UFW firewall.



Check iptables Rules

```
$ sudo iptables -L -n -v
```

Displays iptables firewall rules and policies.



Check Routing Table

```
$ ip route
```

Displays the system routing table.



Check Hosts File

```
$ cat /etc/hosts
```

Displays static host entries.



Check DNS Configuration

```
$ cat /etc/resolv.conf
```

Shows system DNS nameserver settings.



Network Connectivity Test

```
$ ping <ip_or_domain>
```

Tests connectivity to a host.



Traceroute to Destination

```
$ traceroute <ip_or_domain>
```

Traces the network path to a destination.



Check Fail2ban Status

```
$ sudo fail2ban-client status
```

Checks Fail2ban status and active jails.



Analyze Network Connections

```
$ sudo tcpdump -i any
```

Captures and analyzes network packets.



WHY IT MATTERS

Monitoring network activity helps detect unauthorized connections, open ports, and malicious communication with external hosts.



TIP

Regularly review open ports, firewall rules, and active connections to identify anomalies and potential threats early.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

6. LOG MONITORING



View System Logs

```
$ less /var/log/syslog
```

Views general system log file.



View Authentication Logs

```
$ less /var/log/auth.log
```

Views authentication and login related logs.



View Last Login Records

```
$ last
```

Shows last login records of users.



Follow Logs in Real-time

```
$ tail -f /var/log/syslog
```

Monitors logs in real-time.



Search in Log Files

```
$ grep "keyword" \
/var/log/syslog
```

Searches for a keyword in log files.



View Logs by Date

```
$ journalctl --since "2024-05-01"
```

Views logs from a specific date.



View Systemd Journal

```
$ journalctl
```

Displays systemd journal logs.



Search in Journal Logs

```
$ journalctl -k
```

Searches kernel messages in journal.



Failed Login Attempts

```
$ grep "Failed password" \
/var/log/auth.log
```

Finds failed SSH login attempts.



Successful Login Entries

```
$ grep "Accepted password" \
/var/log/auth.log
```

Finds successful login entries.



Sudo Command Usage

```
$ grep "sudo" /var/log/auth.log
```

Shows sudo command usage by users.



Check for Suspicious Activity

```
$ grep -i "error\|denied\|fail" \
/var/log/syslog
```

Finds errors, denied or failed events.



WHY IT MATTERS

Log monitoring helps detect suspicious activities, security incidents, and policy violations early. Always review authentication and system logs regularly.



TIP

Forward important logs to a SIEM for centralized monitoring and correlation. Set up alerts for critical events like failed logins and privilege use.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

7. LINUX HARDENING BASICS



Disable Unnecessary Services

```
$ systemctl list-unit-files
--type=service --state=enabled
```

Lists enabled services. Disable unnecessary services.



Disable Unused User Accounts

```
$ usermod -L username
```

Locks a user account without deleting it.



Remove Unnecessary Packages

```
$ sudo apt autoremove
# or yum autoremove
```

Removes unused dependencies and packages.



Set Strong Password Policy

```
$ sudo nano /etc/login.defs
```

Configure password complexity and expiration policies.



Secure Important Directories

```
$ chmod 750 /etc
$ chmod 750 /root
```

Restrict access to sensitive directories.



Disable Root Login (SSH)

```
$ sudo nano /etc/ssh/sshd_config
# set: PermitRootLogin no
```

Disables direct root login via SSH.



Enable Firewall (UFW)

```
$ sudo ufw enable
$ sudo ufw default deny incoming
```

Enables firewall and denies all incoming connections by default.



Set Secure File Permissions

```
$ chmod 600 /etc/shadow
$ chmod 644 /etc/passwd
```

Ensure critical files have secure permissions.



Enable System Auditing

```
$ sudo apt install auditd
$ sudo systemctl enable auditd
```

Installs and enables auditd for system activity auditing.



Enable Automatic Security Updates

```
$ sudo apt install unattended-upgrades
$ sudo dpkg-reconfigure --priority=low
unattended-upgrades
```

Enables automatic security updates to reduce vulnerabilities.



Enable Kernel Hardening (sysctl)

```
$ sudo nano /etc/sysctl.conf
# Apply hardening settings
$ sudo sysctl -p
```

Apply kernel hardening parameters to improve security.



Check System Security Score

```
$ sudo lynis audit system
```

Run a security audit to find weaknesses and improvements.



WHY IT MATTERS

Hardening reduces the attack surface and makes it harder for attackers to gain access, escalate privileges, or maintain persistence.



TIP

Follow the principle of least privilege. Disable what you don't use, keep systems updated, and monitor continuously.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

8. SSH SECURITY

**Check SSH Service Status**

```
$ systemctl status ssh
```

Checks if SSH service is running.

**Check SSH Service Enablement**

```
$ systemctl is-enabled ssh
```

Checks if SSH service is enabled on boot.

**Check SSH Configuration**

```
$ cat /etc/ssh/sshd_config
```

Displays SSH server configuration file.

**Disable Root Login via SSH**

```
$ sudo nano /etc/ssh/sshd_config  
# set: PermitRootLogin no
```

Disables root login over SSH.

**Disable Password Authentication**

```
$ sudo nano /etc/ssh/sshd_config  
# set: PasswordAuthentication no
```

Disables password login and allows key-based authentication only.

**Change SSH Port**

```
$ sudo nano /etc/ssh/sshd_config  
# set: Port 2222
```

Changes default SSH port (22) to a custom port.

**Allow SSH Access for Specific Users**

```
$ sudo nano /etc/ssh/sshd_config  
# set: AllowUsers user1 user2
```

Restricts SSH access to specified users only.

**Deny SSH Access for Specific Users**

```
$ sudo nano /etc/ssh/sshd_config  
# set: DenyUsers user3 user4
```

Denies SSH access to specified users.

**Check Failed SSH Login Attempts**

```
$ grep "Failed password" \  
/var/log/auth.log
```

Finds failed SSH login attempts.

**Limit SSH Login Attempts**

```
$ sudo nano /etc/ssh/sshd_config  
# set: MaxAuthTries 3
```

Limits number of authentication attempts before disconnect.

**Check Fail2ban Status for SSH**

```
$ sudo fail2ban-client status sshd
```

Checks Fail2ban status for SSH protection.

**Restart SSH Service**

```
$ sudo systemctl restart ssh
```

Applies SSH configuration changes by restarting service.



WHY IT MATTERS

Securing SSH reduces the risk of brute force attacks, unauthorized access, and privilege escalation. Use key-based authentication and restrict access whenever possible.



TIP

Use SSH key authentication, change default port, monitor logs, and implement Fail2ban to protect against brute force attacks.

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ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

9. PACKAGE SECURITY



	List Installed Packages (Debian/Ubuntu)	<code>\$ dpkg -l</code>	Lists all installed packages on Debian/Ubuntu systems.
	List Installed Packages (RHEL/CentOS)	<code>\$ rpm -qa</code>	Lists all installed packages on RHEL/CentOS systems.
	Search Package by Name (Debian/Ubuntu)	<code>\$ apt search package_name</code>	Searches for a package in Debian/Ubuntu repositories.
	Search Package by Name (RHEL/CentOS)	<code>\$ yum search package_name</code> <code># or: dnf search package_name</code>	Searches for a package in RHEL/CentOS repositories.
	Update Package List (Debian/Ubuntu)	<code>\$ sudo apt update</code>	Updates package index on Debian/Ubuntu systems.
	Update Package List (RHEL/CentOS)	<code>\$ sudo yum makecache fast</code> <code># or: sudo dnf makecache</code>	Updates package index on RHEL/CentOS systems.
	Upgrade Installed Packages (Debian/Ubuntu)	<code>\$ sudo apt upgrade -y</code>	Upgrades installed packages on Debian/Ubuntu.
	Upgrade Installed Packages (RHEL/CentOS)	<code>\$ sudo yum update -y</code> <code># or: sudo dnf update -y</code>	Upgrades installed packages on RHEL/CentOS.
	Show Package Information (Debian/Ubuntu)	<code>\$ apt show package_name</code>	Shows detailed information about a package.
	Show Package Information (RHEL/CentOS)	<code>\$ yum info package_name</code> <code># or: dnf info package_name</code>	Shows detailed information about a package.
	Check Package Vulnerabilities (Debian/Ubuntu)	<code>\$ sudo apt install --only-upgrade -s</code> <code># or: sudo apt list --upgradable</code>	Checks for available security updates.
	Check Package Vulnerabilities (RHEL/CentOS)	<code>\$ sudo yum updateinfo list security</code> <code># or: sudo dnf updateinfo list security</code>	Lists available security updates for packages.
	Remove Unused Packages (Debian/Ubuntu)	<code>\$ sudo apt autoremove -y</code>	Removes unused dependencies and packages.
	Remove Unused Packages (RHEL/CentOS)	<code>\$ sudo yum autoremove -y</code> <code># or: sudo dnf autoremove -y</code>	Removes unused dependencies and packages.
	Verify Package Integrity (RPM Based Systems)	<code>\$ rpm -Va</code>	Verifies integrity of installed RPM packages.



WHY IT MATTERS

Keeping packages updated and verifying their integrity helps prevent exploitation of known vulnerabilities and ensures system stability and security.



TIP

Regularly check for package updates and security advisories. Remove unnecessary packages to reduce the attack surface.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

10. SUSPICIOUS ACTIVITY CHECKS



	Check Currently Logged in Users	<pre>\$ who \$ w</pre>	Shows who is currently logged in and what they are doing.
	Check Recent Login History	<pre>\$ last -a</pre>	Shows all recent login attempts (successful and failed).
	Check Failed Login Attempts	<pre>\$ grep "Failed password" \ /var/log/auth.log</pre>	Finds failed SSH or user login attempts.
	Check Successful Login Attempts	<pre>\$ grep "Accepted password" \ /var/log/auth.log</pre>	Finds successful SSH or user login attempts.
	Check Recent Sudo Usage	<pre>\$ grep "sudo" /var/log/auth.log</pre>	Shows recent sudo command usage by users.
	Check Modified Files (Last 24 Hours)	<pre>\$ find / -type f -mtime -1 -ls 2>/dev/null</pre>	Lists files modified in the last 24 hours.
	Check New Files (Last 24 Hours)	<pre>\$ find / -type f -ctime -1 -ls 2>/dev/null</pre>	Lists new files created in the last 24 hours.
	Check Cron Jobs (Users)	<pre>\$ crontab -l -u <username></pre>	Shows cron jobs for a specific user.
	Check System Cron Jobs (Root)	<pre>\$ ls -l /etc/cron* \$ cat /etc/crontab</pre>	Lists system-wide cron jobs and schedules.
	Check Running Processes	<pre>\$ ps aux --sort=-%cpu head -20</pre>	Shows top running processes (sorted by CPU usage).
	Check Suspicious Network Connections	<pre>\$ netstat -tulnp # or: ss -tulnp</pre>	Lists listening ports and active network connections.
	Check Large Log Files (Possible Abuse)	<pre>\$ du -sh /var/log/* sort -hr</pre>	Finds unusually large log files that may indicate issues.
	Check for Rootkits (ClamAV)	<pre>\$ sudo clamscan -r / # or: rkhunter --check</pre>	Scans system for rootkits and malicious files.
	Check for Suspicious Processes (by Name)	<pre>\$ ps aux egrep -i "nc netcat python perl ruby"</pre>	Searches for suspicious processes like reverse shells or scripts.
	Check Kernel Messages for Errors/Alerts	<pre>\$ dmesg tail -50</pre>	Shows last 50 kernel messages for errors or warnings.



WHY IT MATTERS

Regularly checking for suspicious activity helps detect intrusions early, identify attacker behavior, and reduce the impact of security incidents.



TIP

Automate these checks with scripts and alerts. Correlate findings with logs and threat intelligence for faster detection and response.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

11. INTEGRITY MONITORING (5 ESSENTIAL COMMANDS)



#	WHAT IT DOES	COMMAND	WHERE / EXAMPLE
1 	Calculate File Checksum Generates a checksum (hash) for a file to verify its integrity.	<pre>\$ sha256sum /path/to/file</pre>	Where: Local file Example: <pre>\$ sha256sum /etc/passwd</pre>
2 	Calculate Checksum for All Files in Directory Recursively calculates checksums for all files in a directory.	<pre>\$ find /path/to/dir -type f -exec sha256sum {} + > checksums.txt</pre>	Where: Directory Example: <pre>\$ find /var/www -type f -exec sha256sum {} + > checksums.txt</pre>
3 	Verify File Integrity Using Checksum Verifies a file's integrity by comparing with a known checksum.	<pre>\$ echo "<checksum> /path/to/file" sha256sum -c -</pre>	Where: Local file Example: <pre>\$ echo "d2d2... /etc/passwd" sha256sum -c -</pre>
4 	Check File or Directory with AIDE AIDE scans and verifies the integrity of files and directories against its database.	<pre>\$ sudo aide --check \$ sudo aide --check --report</pre>	Where: System-wide Example: <pre>\$ sudo aide --check</pre> (Generates report of changes)
5 	Monitor Real-time File Integrity with inotifywait Monitors file or directory changes in real-time.	<pre>\$ inotifywait -m -r -e modify,create,delete /path/to/dir</pre>	Where: Directory Example: <pre>\$ inotifywait -m -r -e modify,create,delete /etc</pre>



WHY IT MATTERS

Integrity monitoring helps detect unauthorized changes, file tampering, rootkits, and other malicious activities early. It ensures system files remain trustworthy and supports incident detection and response.



TIP

Baseline your critical files regularly and store checksums securely. Automate scans and alert on any unexpected changes.

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

12. LOG ANALYSIS & LOG MONITORING



#	WHAT IT DOES	COMMAND	WHERE / EXAMPLE
1	 View System Log Displays the main system log file.	<pre>\$ cat /var/log/syslog # or \$ less /var/log/syslog</pre>	Where: System-wide Example: <pre>\$ cat /var/log/syslog less</pre>
2	 View Authentication Log Shows login, logout, and authentication events.	<pre>\$ cat /var/log/auth.log # or \$ less /var/log/auth.log</pre>	Where: Debian/Ubuntu Example: <pre>\$ grep "Failed password" /var/log/auth.log</pre>
3	 View Secure Log Shows authentication events (RHEL/CentOS).	<pre>\$ cat /var/log/secure # or \$ less /var/log/secure</pre>	Where: RHEL/CentOS Example: <pre>\$ grep "Failed password" /var/log/secure</pre>
4	 View Kernel Messages Displays kernel log messages.	<pre>\$ dmesg less # or \$ tail -f /var/log/kern.log</pre>	Where: System-wide Example: <pre>\$ dmesg grep -i error</pre>
5	 Check for Errors Filters and shows error messages.	<pre>\$ grep -i error /var/log/syslog</pre>	Where: System-wide Example: <pre>\$ grep -i error /var/log/syslog</pre>
6	 Check for Warnings Filters and shows warning messages.	<pre>\$ grep -i warning /var/log/syslog</pre>	Where: System-wide Example: <pre>\$ grep -i warning /var/log/syslog</pre>
7	 Search Logs for Keyword Searches for a keyword in a log file.	<pre>\$ grep "keyword" /var/log/syslog</pre>	Where: Any log file Example: <pre>\$ grep "sudo" /var/log/auth.log</pre>
8	 Monitor Log in Real-time Continuously monitors a log file.	<pre>\$ tail -f /var/log/syslog</pre>	Where: Any log file Example: <pre>\$ tail -f /var/log/auth.log</pre>
9	 Show Login Events Lists successful login events.	<pre>\$ grep "Accepted password" /var/log/auth.log</pre>	Where: Debian/Ubuntu Example: <pre>\$ grep "Accepted password" /var/log/auth.log</pre>
10	 Show Failed Login Events Lists failed login attempts.	<pre>\$ grep "Failed password" /var/log/auth.log</pre>	Where: Debian/Ubuntu Example: <pre>\$ grep "Failed password" /var/log/auth.log</pre>



WHY IT MATTERS

Logs provide crucial visibility into system activities, user actions, and potential security incidents. Regular log analysis helps in early threat detection, forensics, and compliance.



TIP

Centralize logs using a SIEM or log server for better correlation and alerting. Set up log rotation to manage disk space and ensure logs are not tampered with.

Useful command: `logrotate -d /etc/logrotate.conf`

CHEAT SHEET

ESSENTIAL COMMANDS FOR SOC ANALYSTS & BLUE TEAM

13. INCIDENT RESPONSE QUICK COMMANDS



#	WHAT IT DOES	COMMAND	WHERE / EXAMPLE
1 	Identify Active Connections Shows all active network connections and listening ports.	<pre>\$ ss -tulpen</pre>	Where: System-wide Example: <pre>\$ ss -tulpen less</pre>
2 	Kill Suspicious Process Terminates a process by PID.	<pre>\$ kill -9 <PID></pre>	Where: System-wide Example: <pre>\$ kill -9 12345</pre>
3 	Collect Running Processes Captures a snapshot of currently running processes.	<pre>\$ ps aux > /tmp/ps_snapshot.txt</pre>	Where: System-wide Example: <pre>\$ ps aux > /tmp/ps_snapshot.txt</pre>
4 	Check Disk Usage Identifies disk usage and potential space issues.	<pre>\$ df -h</pre>	Where: System-wide Example: <pre>\$ df -h</pre>
5 	Collect System Info for Analysis Gathers key system information for incident analysis.	<pre>\$ uname -a > /tmp/system_info.txt \$ uptime >> /tmp/system_info.txt \$ who >> /tmp/system_info.txt \$ ip a >> /tmp/system_info.txt</pre>	Where: System-wide Example: <pre>\$ uname -a > /tmp/system_info.txt \$ uptime >> /tmp/system_info.txt \$ who >> /tmp/system_info.txt \$ ip a >> /tmp/system_info.txt</pre>



WHY IT MATTERS

Quick and effective incident response can contain threats, minimize damage, and support faster recovery. These commands help SOC analysts act quickly and collect critical evidence.



TIP

Always document your actions and collected data during an incident. Use a dedicated directory for evidence collection and maintain chain of custody.

Useful command: `script /tmp/incident_session.log`